

Automated Interpretable Neuron Screening for Calcium Imaging

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Abstract—Manual curation of candidate neurons is a persistent bottleneck in one-photon calcium imaging: component-extraction algorithms deliberately over-segment to preserve sensitivity, leaving experts to inspect hundreds to thousands of candidates per session by hand. We present an automated, fully transparent screening system. Redundant components—the same neuron split into several copies—are first removed by a correlation-network step that keeps one representative per group. Each surviving candidate is then described by interpretable features spanning spatial morphology, trace statistics, wavelet-derived calcium-event properties, and reconstruction quality, and classified by an Explainable Boosting Machine under a precision-weighted objective; a cascaded fallback restores event features for neurons on which transient detection fails. Validated by session-stratified cross-validation on a large hippocampal dataset (92,242 candidates from 187 sessions of freely-behaving mice), the system reaches 0.976 precision and 0.933 recall, cutting curation from hours to minutes with a per-feature audit trail for every decision.

Index Terms—calcium imaging, quality control, correlation networks, wavelet event detection, interpretable machine learning, explainable boosting machine

I. INTRODUCTION

One-photon calcium imaging with miniaturized microscopes (miniscopes) records the activity of large neuronal populations in freely behaving animals. Single-neuron signals

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are extracted by constrained matrix-factorization algorithms such as CNMF-E [1], implemented in the CaImAn suite [2]. By design these algorithms favor sensitivity over specificity: they over-segment the field of view so that no genuine neuron is missed, accepting that many false candidates—from motion, neuropil contamination, imaging aberrations, edge effects, and over-splitting of single cells into several copies—are produced alongside real neurons.

The standard workflow therefore ends with a manual step in which an expert inspects every candidate, weighing its spatial footprint, temporal activity, and quality metrics, and labels it a neuron or an artifact. For a typical session of 500–1500 candidates this requires one to three hours, and the implicit criteria that distinguish a “real” neuron are inconsistent between annotators and hard to transfer.

We automate this bottleneck with an interpretable pipeline whose contribution is fourfold: (i) a correlation-network stage that removes duplicate components by clustering co-active footprints; (ii) a 35-dimensional interpretable feature space, including calcium-event descriptors from wavelet analysis; (iii) a glass-box Explainable Boosting Machine (EBM) trained against an asymmetric-cost objective; and (iv) validation at scale—92,242 expert-labeled candidates from 187 sessions—yielding expert-level accuracy with a complete decision audit trail.

II. METHODS

A. Animals, imaging, and preprocessing

The system was developed on data from a large hippocampal imaging program [3]; we summarize the acquisition relevant to quality control. Adult C57BL/6 mice (with 5xFAD animals in a memory-impairment arm) were used. Under anesthesia, 300 nL of AAV1.CAG.GCaMP6s were injected unilaterally into dorsal hippocampal area CA1 (AP -1.94 , ML $+1.46$, DV -1.2 mm); one week later a 1-mm GRIN lens was implanted above CA1, and a baseplate for a miniature microscope (UCLA Miniscope V4.4) was mounted in a third procedure. After recovery, CA1 population activity was imaged in freely moving animals across four free-exploration and contextual-memory paradigms—among them cue-rich open-field exploration in a 44×44 cm arena over four daily 10-min sessions, contextual fear conditioning, a patterned open field, and a repeated open-field memory task. Overhead behavioral video was synchronized with the imaging stream in Bonsai [4]. All procedures were approved by the Bioethics Commission of Lomonosov Moscow State University (application 159-g, meeting 154-d-r, 17 August 2023).

Videos were motion-corrected with NoRMCorre [5] and decomposed into spatial footprints and calcium traces with CNMF-E [1] in the authors' BEARMIND pipeline, built on CaImAn [2]. CNMF-E used estimated neuron size 3–6 px (at $3 \mu\text{m}/\text{px}$), local-correlation threshold 0.85–0.95, and peak-to-noise ratio 3–15, with a first pass keeping components of spatial correlation ≥ 0.95 and SNR ≥ 3 . Traces were normalized as dF/F and footprints registered across sessions with CellReg [6]. These steps yield **92,242 candidate components from 187 sessions**, each manually labeled as a genuine neuron (81.1%) or an artifact (18.9%)—the decisions our system learns to reproduce.

B. Network-based duplicate removal

CNMF-E frequently splits one physical neuron into several near-identical copies. We remove this redundancy with a correlation-network step. Treating candidates as nodes, we connect two components when their dF/F traces are strongly Spearman-correlated ($\rho \geq 0.4$) reproducibly across temporal blocks of the recording. The connected components of the resulting graph are groups of putative duplicates, ranked by their mean internal correlation (a graph-density score). Within each group, components whose footprints are spatially adjacent are taken to be the same cell: the highest-SNR copy is retained and the others merged or discarded. The spatial-adjacency requirement guards against collapsing genuinely distinct but co-active neurons, which are correlated yet spatially separated. Casting deduplication as clustering of a correlation graph removes redundancy before classification and links candidate screening to the analysis of correlation networks.

C. Interpretable feature space

Each surviving candidate is described by 35 features in six groups: *spatial/morphological* (11; area, circularity, convexity, aspect ratio, eccentricity, edge distance, local density, ...),

temporal/trace (9; skewness, kurtosis, baseline drift, noise, bimodality, Hurst exponent, ...), *event-based* (8; rate, fit quality, signal-to-noise, rise/decay times, amplitude variability, half-crossing rate), *reconstruction* (4; R^2 , NMAE, NRMSE, residual SNR of a calcium-kinetics fit), *extractor-provided* (2; CaImAn SNR and spatial-correlation score), and a *kinetics-success flag* (1; see Sec. II-E). No single metric separates neurons from artifacts; the discriminative power is distributed across these complementary views.

D. Wavelet-based event characterization

The most discriminative features describe calcium transients, which we detect in the time–scale plane with a wavelet-ridge method [7] implemented in our DRIADA toolkit [8]. The dF/F trace is smoothed and mapped by a continuous wavelet transform using generalized Morse wavelets $\Psi_{\beta,\gamma}(\omega) \propto \omega^\beta e^{-\omega^\gamma}$ with $(\beta, \gamma) = (2, 3)$ —an analytic, low-oscillation wavelet well matched to the asymmetric rise and decay of calcium events; scales span transient durations of 0.275–2.25 s. Local maxima of the transform are linked across scales into ridges, and a ridge is accepted as an event when it is sufficiently long, strong, and slow (thresholds on ridge length, peak magnitude, maximal scale, and duration). From the detected events E we derive descriptors such as event rate, amplitude variability, rise/decay times, an event signal-to-noise ratio, and the event-restricted reconstruction quality

$$R_{\text{ev}}^2 = 1 - \frac{\sum_{t \in E} (x_t - \hat{x}_t)^2}{\sum_{t \in E} (x_t - \bar{x}_E)^2}, \quad (1)$$

measured against a single-compartment kinetics fit $\hat{x}$ of the trace x .

E. Cascading kinetics estimation

Ridge-based detection fails on 40–60% of candidates (low SNR, sparse firing, nonstationary baseline), leaving eight event features undefined. A five-tier cascade restores them: (i) wavelet detection with strict criteria; (ii) wavelet with relaxed thresholds (≥ 3 events, $R_{\text{ev}}^2 \geq 0.6$); (iii)–(iv) a robust amplitude-threshold detector at strict then relaxed criteria; and (v) physically plausible defaults ($t_{\text{rise}} = 0.25$ s, $t_{\text{off}} = 2.0$ s). Whether kinetics estimation succeeded is recorded as a feature, letting the classifier discount low-reliability event evidence. Coverage rises from 42% (single attempt) to 80%, and spatial, trace, and reconstruction features remain defined even at tier (v) (graceful degradation).

F. Classifier and decision objective

Candidates are classified by an Explainable Boosting Machine (EBM) [9], [10], a generalized additive model

$$g(\mathbb{E}[y | \mathbf{x}]) = \beta_0 + \sum_j f_j(x_j) + \sum_{(j,k) \in \mathcal{I}} f_{jk}(x_j, x_k), \quad (2)$$

with per-feature shape functions f_j and $|\mathcal{I}| = 20$ learned pairwise interactions. Unlike black-box ensembles, every prediction decomposes exactly into additive feature contributions, yielding a complete audit trail; its accuracy is benchmarked

TABLE I
TEST-SPLIT COMPARISON (THRESHOLD TUNED ON TRAINING DATA;
 $\beta = 0.577$).

Method	Prec.	Rec.	F_β	AUC
CaImAn metrics (best rule)	0.836	0.974	0.867	—
Logistic regression	0.965	0.912	0.951	0.946
Random forest	0.939	0.975	0.948	0.960
EBM (ours)	0.968	0.938	0.960	0.966

against black-box baselines in Sec. III. Because errors are asymmetric—a false positive (artifact kept) can corrupt population statistics, whereas a false negative merely costs statistical power—we optimize

$$F_\beta = (1 + \beta^2) \frac{\text{precision} \cdot \text{recall}}{\beta^2 \text{precision} + \text{recall}}, \quad (3)$$

with $\beta = \sqrt{1/3} \approx 0.577$ [11], weighting false positives three times more than false negatives. Hyperparameters were chosen by stratified cross-validation over 128 configurations. At deployment, candidates may be filtered by transparent rules, by EBM score at a tunable threshold, or by a hybrid that applies rules first and the EBM to the remainder.

III. RESULTS

A. Classification performance

We evaluate by cross-validation stratified by *recording session*, so that candidates from one session never appear in both training and test folds; this precludes exploiting session-specific imaging conditions. Across folds (mean $\pm$ s.d.) the production model attains $F_\beta = 0.965 \pm 0.004$, precision 0.976 ± 0.003 , recall 0.933 ± 0.007 , and AUC 0.978 ± 0.005 . A precision of 97.6% means only 2.4% of accepted candidates are artifacts; recall of 93.3% means 6.7% of genuine neurons are missed—a trade-off deliberately favoring precision. The low cross-session variance confirms stable generalization across paradigms. Because the same neurons recur across days within an animal, we also re-evaluated under a stricter *animal-stratified* split that shares no animal (hence no cross-day-tracked neuron) between train and test: a random-forest probe showed no degradation (F_β $0.948 \rightarrow 0.959$, AUC $0.960 \rightarrow 0.965$), so the reported accuracy reflects cross-animal generalization rather than memorization of recurring neurons.

CaImAn’s own spatial-correlation and SNR metrics are weak discriminators on one-photon data (AUC 0.59 and 0.56 individually); even an optimally tuned threshold on both (Table I, top row) reaches only $F_\beta = 0.87$, no better than indiscriminately keeping every candidate, which is why multi-feature screening is needed. Among learned classifiers (Table I, held-out test split), off-the-shelf logistic regression and random forests reach $F_\beta \approx 0.95$, so the task is learnable; the glass-box EBM matches or exceeds both ($F_\beta = 0.96$, AUC 0.97, highest precision) while remaining fully interpretable—interpretability here costs no accuracy.

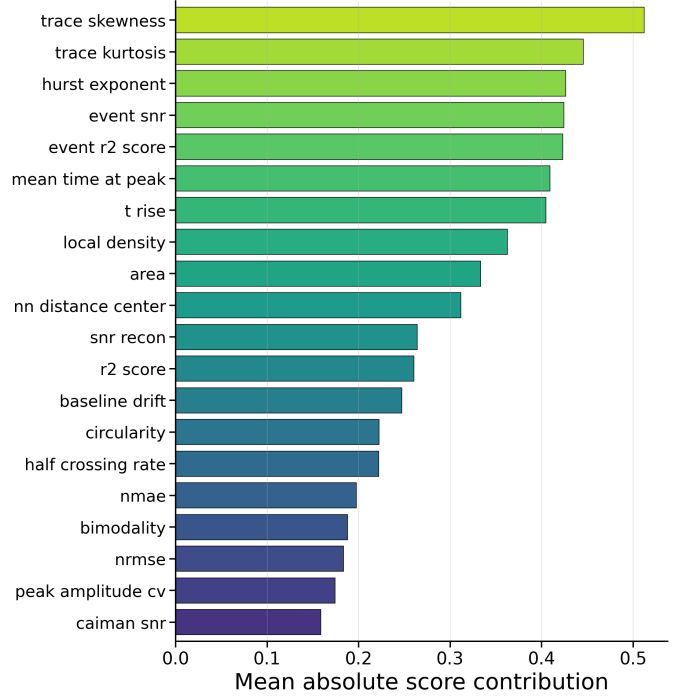


Fig. 1. Mean absolute score contribution of the top features in the production EBM. Temporal trace statistics (skewness, kurtosis) and the Hurst exponent dominate, followed by wavelet event descriptors (event SNR, event R^2); the extractor-provided CaImAn SNR ranks lowest.

B. Which features matter

Feature-importance analysis (Fig. 1) shows that temporal trace statistics carry the strongest discriminative power: trace skewness and kurtosis rank first and second, validating the heavy-tailed, “spiky” signature of genuine calcium dynamics. Wavelet-derived event descriptors rank immediately below, whereas the extractor’s own amplitude SNR ranks lowest—temporal structure is far more informative than raw amplitude.

C. Tuning the operating point

Because the EBM emits a calibrated score, a single threshold tunes the precision/recall balance (Fig. 2). The unconstrained F_β optimum lies near a threshold of 0.66; the deployed operating point (0.72) sits slightly above it, trading a little recall for additional precision, consistent with the asymmetric-cost objective. Lower thresholds (0.3–0.4) maximize recall for exploratory screening, higher thresholds for publication-grade datasets.

D. Practical impact

Manual curation of a 1000-candidate session takes one to two hours; duplicate removal, feature extraction, and classification together process the same session in two to three minutes on a standard workstation—a 30–40 $\times$ speedup enabling same-day analysis. Identical criteria applied to every candidate make the output reproducible, and the additive decomposition of Eq. 2 shows exactly why each decision was made.

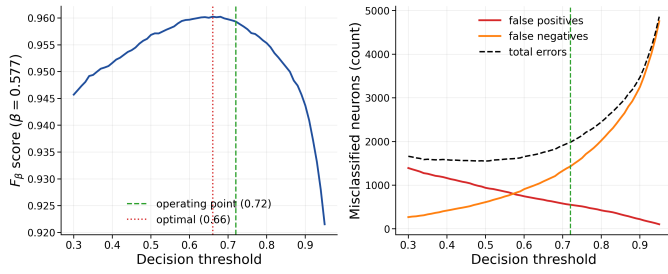


Fig. 2. Threshold analysis on the held-out test split. Left: F_β ($\beta=0.577$) versus decision threshold, with the deployed operating point (0.72) and the F_β optimum (0.66) marked. Right: false positives, false negatives, and total errors versus threshold.

IV. DISCUSSION AND CONCLUSION

We have presented an automated quality-control system that automates the principal manual bottleneck in calcium-imaging preprocessing. Removing duplicates by clustering a correlation network (Sec. II-B) and characterizing activity through wavelet-detected events (Sec. II-D) supply the features that prove most discriminative. Interpretability also enables iterative improvement: the additive structure of Eq. 2 exposes which features drove a decision, so a corrected label becomes a training example for the next round. Across iterations, cross-validated F_β improved from 0.961 to 0.965 and AUC from 0.967 to 0.978, with the largest gains on initially hard cases.

Limitations remain: the system was trained on one-photon GCaMP recordings of hippocampal neurons, so transfer to other modalities or cell types may require retraining; entirely silent neurons are inherently ambiguous; and $\sim 20\%$ of candidates still rely on default kinetics. Natural extensions include tighter coupling to upstream extractors, real-time assessment during acquisition, and session-specific calibration. The broader point is practical: an interpretable model here reaches the accuracy of black-box classifiers while exposing the reason for every decision, and, combined with CNMF-E extraction, reduces curation to reviewing a small set of flagged borderline cases.

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